

Problem A. Number of Pairs

Time Limit 2000 ms

Mem Limit 262144 kB

You are given an array a of n integers. Find the number of pairs (i, j) ($1 \leq i < j \leq n$) where the sum of $a_i + a_j$ is greater than or equal to l and less than or equal to r (that is, $l \leq a_i + a_j \leq r$).

For example, if $n = 3$, $a = [5, 1, 2]$, $l = 4$ and $r = 7$, then two pairs are suitable:

- $i = 1$ and $j = 2$ ($4 \leq 5 + 1 \leq 7$);
- $i = 1$ and $j = 3$ ($4 \leq 5 + 2 \leq 7$).

Input

The first line contains an integer t ($1 \leq t \leq 10^4$). Then t test cases follow.

The first line of each test case contains three integers n, l, r ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq l \leq r \leq 10^9$) — the length of the array and the limits on the sum in the pair.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n overall test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the number of index pairs (i, j) ($i < j$), such that $l \leq a_i + a_j \leq r$.

Examples

Input	Output
4 3 4 7 5 1 2 5 5 8 5 1 2 4 3 4 100 1000 1 1 1 1 5 9 13 2 5 5 1 1	2 7 0 1

Problem B. Don't be late

Time Limit 2000 ms

Problem Statement

Takahashi is meeting up with Aoki.

They have planned to meet at a place that is D meters away from Takahashi's house in T minutes from now.

Takahashi will leave his house now and go straight to the place at a speed of S meters per minute.

Will he arrive in time?

Constraints

- $1 \leq D \leq 10000$
- $1 \leq T \leq 10000$
- $1 \leq S \leq 10000$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

D T S

Output

If Takahashi will reach the place in time, print **Yes**; otherwise, print **No**.

Sample 1

Input	Output
1000 15 80	Yes

It takes 12.5 minutes to go 1000 meters to the place at a speed of 80 meters per minute. They have planned to meet in 15 minutes so he will arrive in time.

Sample 2

Input	Output
2000 20 100	Yes

It takes 20 minutes to go 2000 meters to the place at a speed of 100 meters per minute. They have planned to meet in 20 minutes so he will arrive just on time.

Sample 3

Input	Output
10000 1 1	No

He will be late.

Problem C. Subarray Sums I

Time Limit 1000 ms

Mem Limit 524288 kB

Given an array of n positive integers, your task is to count the number of subarrays having sum x .

Input

The first input line has two integers n and x : the size of the array and the target sum x .

The next line has n integers $a_1, a_2, \dots, a_n$: the contents of the array.

Output

Print one integer: the required number of subarrays.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x, a_i \leq 10^9$

Example

Input	Output
5 7 2 4 1 2 7	3

Problem D. Books

Time Limit 2000 ms

Mem Limit 262144 kB

Input File stdin

Output File stdout

When Valera has got some free time, he goes to the library to read some books. Today he's got t free minutes to read. That's why Valera took n books in the library and for each book he estimated the time he is going to need to read it. Let's number the books by integers from 1 to n . Valera needs a_i minutes to read the i -th book.

Valera decided to choose an arbitrary book with number i and read the books one by one, starting from this book. In other words, he will first read book number i , then book number $i + 1$, then book number $i + 2$ and so on. He continues the process until he either runs out of the free time or finishes reading the n -th book. Valera reads each book up to the end, that is, he doesn't start reading the book if he doesn't have enough free time to finish reading it.

Print the maximum number of books Valera can read.

Input

The first line contains two integers n and t ($1 \leq n \leq 10^5$; $1 \leq t \leq 10^9$) — the number of books and the number of free minutes Valera's got. The second line contains a sequence of n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^4$), where number a_i shows the number of minutes that the boy needs to read the i -th book.

Output

Print a single integer — the maximum number of books Valera can read.

Examples

Input	Output
4 5 3 1 2 1	3

Input	Output
3 3 2 2 3	1

Problem E. Interesting drink

Time Limit 2000 ms

Mem Limit 262144 kB

Vasiliy likes to rest after a hard work, so you may often meet him in some bar nearby. As all programmers do, he loves the famous drink "Beecola", which can be bought in n different shops in the city. It's known that the price of one bottle in the shop i is equal to x_i coins.

Vasiliy plans to buy his favorite drink for q consecutive days. He knows, that on the i -th day he will be able to spend m_i coins. Now, for each of the days he wants to know in how many different shops he can buy a bottle of "Beecola".

Input

The first line of the input contains a single integer n ($1 \leq n \leq 100\,000$) — the number of shops in the city that sell Vasiliy's favourite drink.

The second line contains n integers x_i ($1 \leq x_i \leq 100\,000$) — prices of the bottles of the drink in the i -th shop.

The third line contains a single integer q ($1 \leq q \leq 100\,000$) — the number of days Vasiliy plans to buy the drink.

Then follow q lines each containing one integer m_i ($1 \leq m_i \leq 10^9$) — the number of coins Vasiliy can spend on the i -th day.

Output

Print q integers. The i -th of them should be equal to the number of shops where Vasiliy will be able to buy a bottle of the drink on the i -th day.

Examples

Input	Output
5 3 10 8 6 11 4 1 10 3 11	0 4 1 5

Note

On the first day, Vasiliy won't be able to buy a drink in any of the shops.

On the second day, Vasiliy can buy a drink in the shops 1, 2, 3 and 4.

On the third day, Vasiliy can buy a drink only in the shop number 1.

Finally, on the last day Vasiliy can buy a drink in any shop.

Problem F. Apartments

Time Limit 1000 ms

Mem Limit 524288 kB

There are n applicants and m free apartments. Your task is to distribute the apartments so that as many applicants as possible will get an apartment.

Each applicant has a desired apartment size, and they will accept any apartment whose size is close enough to the desired size.

Input

The first input line has three integers n , m , and k : the number of applicants, the number of apartments, and the maximum allowed difference.

The next line contains n integers $a_1, a_2, \dots, a_n$: the desired apartment size of each applicant. If the desired size of an applicant is x , they will accept any apartment whose size is between $x - k$ and $x + k$.

The last line contains m integers $b_1, b_2, \dots, b_m$: the size of each apartment.

Output

Print one integer: the number of applicants who will get an apartment.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $0 \leq k \leq 10^9$
- $1 \leq a_i, b_i \leq 10^9$

Example

Input	Output
4 3 5 60 45 80 60 30 60 75	2

Problem G. Vacation Together

Time Limit 2000 ms

Problem Statement

There are N people numbered 1 to N .

You are given their schedule for the following D days. The schedule for person i is represented by a string S_i of length D . If the j -th character of S_i is **o**, person i is free on the j -th day; if it is **x**, they are occupied that day.

From these D days, consider choosing some **consecutive** days when all the people are free. How many days can be chosen at most? If no day can be chosen, report 0.

Constraints

- $1 \leq N \leq 100$
- $1 \leq D \leq 100$
- N and D are integers.
- S_i is a string of length D consisting of **o** and **x**.

Input

The input is given from Standard Input in the following format:

```
N D
S1
S2
⋮
SN
```

Output

Print the maximum number of days that can be chosen, or **0** if no day can be chosen.

Sample 1

Input	Output
3 5 X000X 000XX 000X0	2

All the people are free on the second and third days, so we can choose them.
Choosing these two days will maximize the number of days among all possible choices.

Sample 2

Input	Output
3 3 0X0 0X0 0X0	1

Note that the chosen days must be consecutive. (All the people are free on the first and third days, so we can choose either of them, but not both.)

Sample 3

Input	Output
3 3 00X 0X0 X00	0

Print 0 if no day can be chosen.

Sample 4

Input	Output
1 7 0000000	7

Sample 5

Input	Output
5 15 0X0000000000000000 0X00X00000000000X 0X000000000000000X 0XXX000000X000X 0X00000000X000X	5

Problem H. Peak

Time Limit 2000 ms

Problem Statement

Takahashi has placed N gifts on a number line. The i -th gift is placed at coordinate A_i .

You will choose a half-open interval $[x, x + M)$ of length M on the number line and acquire all the gifts included in it.

More specifically, you acquire gifts according to the following procedure.

- First, choose one real number x .
- Then, acquire all the gifts whose coordinates satisfy $x \leq A_i < x + M$.

What is the maximum number of gifts you can acquire?

Constraints

- All input values are integers.
- $1 \leq N \leq 3 \times 10^5$
- $1 \leq M \leq 10^9$
- $0 \leq A_i \leq 10^9$

Input

The input is given from Standard Input in the following format:

```
N M
A_1 A_2 ... A_N
```

Output

Print the answer as an integer.

Sample 1

Input	Output
8 6 2 3 5 7 11 13 17 19	4

For example, specify the half-open interval $[1.5, 7.5)$.

In this case, you can acquire the four gifts at coordinates 2, 3, 5, 7, the maximum number of gifts that can be acquired.

Sample 2

Input	Output
10 1 3 1 4 1 5 9 2 6 5 3	2

There may be multiple gifts at the same coordinate.

Sample 3

Input	Output
10 998244353 100000007 0 1755647 998244353 495 1000000000 1755648 503 1755649 998244853	7

Problem I. Factory Machines

Time Limit 1000 ms

Mem Limit 524288 kB

A factory has n machines which can be used to make products. Your goal is to make a total of t products.

For each machine, you know the number of seconds it needs to make a single product. The machines can work simultaneously, and you can freely decide their schedule.

What is the shortest time needed to make t products?

Input

The first input line has two integers n and t : the number of machines and products.

The next line has n integers $k_1, k_2, \dots, k_n$: the time needed to make a product using each machine.

Output

Print one integer: the minimum time needed to make t products.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq t \leq 10^9$
- $1 \leq k_i \leq 10^9$

Example

Input	Output
3 7 3 2 5	8

Explanation: Machine 1 makes two products, machine 2 makes four products and machine 3 makes one product.

Problem J. Sum of Two Values

Time Limit 1000 ms

Mem Limit 524288 kB

You are given an array of n integers, and your task is to find two values (at distinct positions) whose sum is x .

Input

The first input line has two integers n and x : the array size and the target sum.

The second line has n integers $a_1, a_2, \dots, a_n$: the array values.

Output

Print two integers: the positions of the values. If there are several solutions, you may print any of them. If there are no solutions, print **IMPOSSIBLE**.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x, a_i \leq 10^9$

Example

Input	Output
4 8 2 7 5 1	2 4